

Barocycler® NEP2320

Pressure Cycling Technology Sample Preparation System (PCT SPS)

...the Power of PCT

Pressure Cycling Technology (PCT) uses rapid cycles of hydrostatic pressure between ambient and ultra high levels to control biomolecular interactions, allowing for a high degree of safety, speed, reproducibility, and convenience. This unique, patented technology offers the potential for broad applications in a number of established and emerging fields, including genomic and proteomic sample preparation, pathogen inactivation, drug discovery and development, protein purification, and immunodiagnostics.

...the PCT Sample Preparation System (PCT SPS)

Sample preparation is a significant bottleneck to discoveries in genomic and proteomic research. To address this problem, the PCT SPS was developed. The PCT SPS offers safe, rapid, and reproducible extraction of DNA, RNA, proteins, and small molecules from a wide variety of cells and tissues, particularly those considered "hard-to-lyse". The PCT SPS uses a small, semi-automated bench top instrument (Barocycler NEP2320 or NEP3229), together with single-use sample processing containers (PULSE Tubes).

...the Barocycler NEP2320



...PULSE® Tubes



The compact and versatile Barocycler NEP2320 fills an important need for all laboratories that want improved sample preparation in genomics, proteomics, transcriptomics, and metabolomics. The NEP2320 is a safe and versatile sample preparation instrument designed to use a minimum amount of laboratory bench space, while still providing...*the Power of PCT*. The mechanisms of action of the PCT SPS differ significantly from other sample preparation methods. Specifically, lysis occurs when samples are subjected to alternating cycles of ambient and high pressure, up to 35,000 psi, in the hydrostatic reaction chamber of the Barocycler. The NEP2320 uses available laboratory compressed air, bottled compressed air, or a stand-alone compressor to create high and ultra-high hydrostatic pressure inside the Barocycler reaction chamber. Exquisite control of the PCT process is accomplished by specialized components, a programmable controller, and software. The NEP2320 quickly and efficiently processes one sample at a time. Although smaller than the NEP3229, the NEP2320 still provides the user with many of the features of the Barocycler NEP3229, including temperature control.

The Barocycler NEP2320 uses the same PULSE Tubes (Pressure Used to Lyse Samples for Extraction) as the NEP3229. PULSE Tubes transmit the Power of PCT from the Barocycler instrument to the sample. For bio-molecule extraction, the specimen (such as tissue) is placed on the Lysis Disk in the sample chamber of the PULSE Tube. The Ram is set in place, and processing buffer is added to the PULSE Tube's fluid retention chamber. Next, the PULSE Tube is placed in the reaction chamber of the Barocycler NEP2320, submerging it in the chamber fluid, which is distilled water. Then the upper limit pressure levels are set, the number of pressure cycles is entered, and the time at pressure fixed. As pressure increases, the Ram pushes the specimen from the Sample Chamber through the Lysis Disk and into the Fluid Retention Chamber. When pressure is released, the sample (now partially homogenized) is pulled back through the Lysis Disk by the receding Ram. The combination of physical passage through the Lysis Disk, rapid pressure changes, and other biophysical mechanisms breaks up the cells and tissues of the specimen to quickly and efficiently release nucleic acids, proteins, and small molecules.

Specifications of the PCT SPS*

Barocycler NEP2320			
Maximum Pressure	35 kpsi, 235MPa	Protocol Settings	Pressure Holding Time (up or down) protocol number
Working Pressure Range	5 to 35 kpsi	Maximum number of programmable cycles	99
Chamber construction	Stainless steel	Temp. Range	4 to 37°C (Temperature control not provided)
Pressure Accuracy	+/-2 kpsi	Minimum air pressure needed	90psi
Min suggested air compressor capacity / tank size**	>2 cfm @ 90psi; >2 gal	Number of PULSE Tubes per Chamber	1
Typical Pressurization Time to 35ksi	< 5 sec	Dimensions	19 × 17 × 19 in
Depressurization Time	< 1 sec	Weight	80 lbs
Power Requirement***	120 VAC, 2 Amps, 60 Hz	Laboratory Environment	15 to 30°C, < 85% Humidity
PULSE Tube FT500			
Dimensions	13 mm diameter × 51 mm long	Compatible solvent	Solvent that is compatible with polypropylene and silicon rubber
Material	Tube, cap, ram: Polypropylene O-ring Seal: silicon rubber	Operational Temperature	4 to 37°C
Sample Size****	Solid: 50 – 500 mg Liquid: 1.2 – 1.5 mL	Storage Temperature	- 70 to 35°C

*Specifications may be changed without prior notification

**Faster ramp time is achieved by increasing capacity of air supply

***For US model - can be modified to comply with non-US power requirements

****The combined volume of solid sample and lysis buffer needs to be 1.2 – 1.5 mL

Features & Benefits of the PCT SPS

- **Safe** PULSE Tubes offer a closed system to reduce sample handling and minimize exposure to pathogens/toxins
- **Fast** Nucleic acids, proteins, and small molecules are rapidly released from a wide variety of cells and tissues
- **Powerful** Up to 35,000 PSI (235 MPa, 2.35 kbar) can be used to lyse samples and release excellent quality and quantity of bio-molecules
- **Efficient** High quantity of high quality bio-molecules quickly and easily released
- **Versatile** Animal, plant, and microbial samples can be processed; either standard or user defined protocols can be used
- **Reproducible** Computer controlled protocols mean consistent extraction each time, every time

Publications

Schumacher, R.T., et al., (2002). An Automated Sample Preparation Solution for Nucleic Acid and Protein Extraction from Cells and Tissues. *Am. Laboratory*, 34, 38-43

Smejkal, G.B. et al., (2007). "Sample Preparation for Two-Dimensional Gel Electrophoresis Using Pressure Cycling Technology" *Analytical Biochemistry*. Vol. 363, Issue 2, 309-311

Okubara, P. et al., (2007). "Pressure Cycling Technology Sample Preparation Systems (PCT SPS) Improves Quantification of Pathogen DNA in Plants and Soil" Poster Presentation. Plant and Animal Genomics Meeting. San Diego, CA

Tisa, L. et al., (2007). "Pressure Cycling Technology (PCT) Facilitates Analysis of Frankia Proteome" Poster Presentation. US HUPO. Seattle, WA

Calvert, V.S., et al., (2006). "Signal Pathway Proteomic Analysis of Human Adipose Tissue Using Phosphoprotein Arrays Insights into the Pathophysiology of Nonalcoholic Fatty Liver Disease (NAFLD) and Type 2 Diabetes (DM). Presentation. US HUPO.



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